Language Models as Knowledge Bases?

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Abstract

Recent progress in pretraining language models on large textual corpora led to a surge of improvements for downstream NLP tasks. Whilst learning linguistic knowledge, these models may also be storing relational knowledge present in the training data, and may be able to answer queries structured as "fill-in-the-blank" cloze statements. Language models have many advantages over structured knowledge bases: they require no schema engineering, allow practitioners to query about an open class of relations, are easy to extend to more data, and require no human supervision to train. We present an in-depth analysis of the relational knowledge already present (without fine-tuning) in a wide range of state-of-the-art pretrained language models. We find that (i) without fine-tuning, BERT contains relational knowledge competitive with traditional NLP methods that have some access to oracle knowledge, (ii) BERT also does remarkably well on open-domain question answering against a supervised baseline, and (iii) certain types of factual knowledge are learned much more readily than others by standard language model pretraining approaches. The surprisingly strong ability of these models to recall factual knowledge without any fine-tuning demonstrates their potential as unsupervised open-domain QA systems. The code to reproduce our analysis is available at `https://github.com/facebookresearch/LAMA`.

1 Introduction

Recently, pretrained high-capacity language models such as ELMo (Peters et al., 2018a) and BERT (Devlin et al., 2018a) have become increasingly important in NLP. They are optimised to either predict the next word in a sequence or some masked word anywhere in a given sequence (*e.g.* "Dante was born in [Mask] in the year 1265."). The parameters of these models appear to store vast amounts of linguistic knowledge (Peters et al., 2018b; Goldberg, 2019; Tenney et al., 2019) useful for downstream tasks. This knowledge is usually accessed either by conditioning on latent context representations produced by the original model or by using the original model weights to initialize a task-specific model which is then further fine-tuned. This type of knowledge transfer is crucial for current state-of-the-art results on a wide range of tasks.

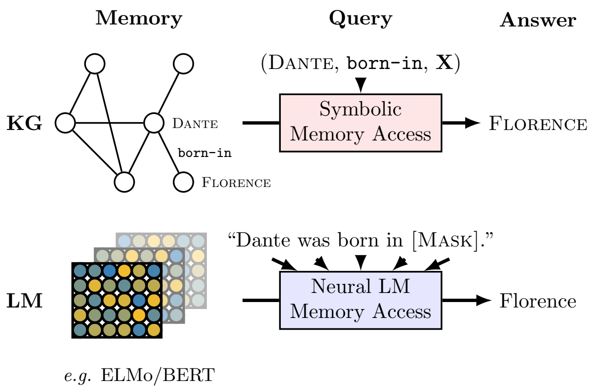

Figure 1: Querying knowledge bases (KB) and language models (LM) for factual knowledge.

In contrast, knowledge bases are effective solutions for accessing annotated gold-standard relational data by enabling queries such as (Dante, `born-in`, **X**). However, in practice we often need to *extract* relational data from text or other modalities to populate these knowledge bases. This requires complex NLP pipelines involving entity extraction, coreference resolution, entity linking and relation extraction (Surdeanu and Ji, 2014)—components that often need supervised data and fixed schemas. Moreover, errors can easily propagate and accumulate throughout the pipeline. Instead, we could attempt to query neural language models for relational data by asking them to fill in masked tokens in sequences like "Dante was born in [MASK]", as illustrated in Figure 1. In this setting, language models come with various attractive properties: they require no schema engineering, do not need human annotations, and they support an open set of queries.

Given the above qualities of language models as potential representations of relational knowledge, we are interested in the relational knowledge already present in pretrained *off-the-shelf* language models such as ELMo and BERT. How much relational knowledge do they store? How does this differ for different types of knowledge such as facts about entities, common sense, and general question answering? How does their performance without fine-tuning compare to symbolic knowledge bases automatically extracted from text? Beyond gathering a better general understanding of these models, we believe that answers to these questions can help us design better unsupervised knowledge representations that could transfer factual and commonsense knowledge reliably to downstream tasks such as commonsense (visual) question answering (Zellers et al., 2018; Talmor et al., 2019) or reinforcement learning (Branavan et al., 2011; Chevalier-Boisvert et al., 2018; Bahdanau et al., 2019; Luketina et al., 2019).

For the purpose of answering the above questions we introduce the *LAMA* (LAnguage Model Analysis) probe, consisting of a set of knowledge sources, each comprised of a set of facts. We define that a pretrained language model *knows* a fact (SUBJECT, `relation`, OBJECT) such as (DANTE, `born-in`, FLORENCE) if it can successfully predict masked objects in cloze sentences such as "Dante was born in ____" expressing that fact. We test for a variety of types of knowledge: relations between entities stored in Wikidata, common sense relations between concepts from ConceptNet, and knowledge necessary to answer natural language questions in SQuAD. In the latter case we manually map a subset of SQuAD questions to cloze sentences.

Our investigation reveals that (i) the largest BERT model from Devlin et al. (2018b) (BERT-large) captures (accurate) relational knowledge comparable to that of a knowledge base extracted with an off-the-shelf relation extractor and an oracle-based entity linker from a corpus known to express the relevant knowledge, (ii) factual knowledge can be recovered surprisingly well from pretrained language models, however, for some relations (particularly N-to-M relations) performance is very poor, (iii) BERT-large consistently outperforms other language models in recovering factual and commonsense knowledge while at the same time being more robust to the phrasing of a query, and (iv) BERT-large achieves remarkable results for open-domain QA, reaching 57.1% precision@10 compared to 63.5% of a knowledge base constructed using a task-specific supervised relation extraction system.

2 Background

In this section we provide background on language models. Statistics for the models that we include in our investigation are summarized in Table 1.

2.1 Unidirectional Language Models

Given an input sequence of tokens $\mathbf{w} = [w_1, w_2, \ldots, w_N]$, unidirectional language models commonly assign a probability $p(\mathbf{w})$ to the sequence by factorizing it as follows

$$p(\mathbf{w}) = \prod_t p(w_t \mid w_{t-1}, \ldots, w_1). \quad (1)$$

A common way to estimate this probability is using neural language models (Mikolov and Zweig, 2012; Melis et al., 2017; Bengio et al., 2003) with

$$p(w_t \mid w_{t-1}, \ldots, w_1) = \mathrm{softmax}(\mathbf{W}\mathbf{h}_t + \mathbf{b}) \quad (2)$$

where $\mathbf{h}_t \in \mathbb{R}^k$ is the output vector of a neural network at position t and $\mathbf{W} \in \mathbb{R}^{|\mathcal{V}| \times k}$ is a learned parameter matrix that maps $\mathbf{h}_t$ to unnormalized scores for every word in the vocabulary $\mathcal{V}$. Various neural language models then mainly differ in how they compute $\mathbf{h}_t$ given the word history, *e.g.*, by using a multi-layer perceptron (Bengio et al., 2003; Mikolov and Zweig, 2012), convolutional layers (Dauphin et al., 2017), recurrent neural networks (Zaremba et al., 2014; Merity et al., 2016; Melis et al., 2017) or self-attention mechanisms (Radford et al., 2018; Dai et al., 2019; Radford et al., 2019).

fairseq-fconv: Instead of commonly used recurrent neural networks, Dauphin et al. (2017) use multiple layers of gated convolutions. We use the pretrained model in the fairseq[1] library in our study. It has been trained on the WikiText-103 corpus introduced by Merity et al. (2016).

[1] `https://github.com/pytorch/fairseq`

Model	Base Model	#Parameters	Training Corpus	Corpus Size
fairseq-fconv (Dauphin et al., 2017)	ConvNet	324M	WikiText-103	103M Words
Transformer-XL (large) (Dai et al., 2019)	Transformer	257M	WikiText-103	103M Words
ELMo (original) (Peters et al., 2018a)	BiLSTM	93.6M	Google Billion Word	800M Words
ELMo 5.5B (Peters et al., 2018a)	BiLSTM	93.6M	Wikipedia (en) & WMT 2008-2012	5.5B Words
BERT (base) (Devlin et al., 2018a)	Transformer	110M	Wikipedia (en) & BookCorpus	3.3B Words
BERT (large) (Devlin et al., 2018a)	Transformer	340M	Wikipedia (en) & BookCorpus	3.3B Words

Table 1: Language models considered in this study.

Transformer-XL: Dai et al. (2019) introduce a large-scale language model based on the Transformer (Vaswani et al., 2017). Transformer-XL can take into account a longer history by caching previous outputs and by using relative instead of absolute positional encoding. It achieves a test perplexity of 18.3 on the WikiText-103 corpus.

2.2 Bidirectional "Language Models"[2]

So far, we have looked at language models that predict the next word given a history of words. However, in many downstream applications we mostly care about having access to contextual representations of words, *i.e.*, word representations that are a function of the entire context of a unit of text such as a sentence or paragraph, and not only conditioned on previous words. Formally, given an input sequence $\mathbf{w} = [w_1, w_2, \ldots, w_N]$ and a position $1 \leq i \leq N$, we want to estimate $p(w_i) = p(w_i \mid w_1, \ldots, w_{i-1}, w_{i+1}, \ldots, w_N)$ using the left and right context of that word.

ELMo: To estimate this probability, Peters et al. (2018a) propose running a forward and backward LSTM (Hochreiter and Schmidhuber, 1997), resulting in $\overrightarrow{\mathbf{h}}_i$ and $\overleftarrow{\mathbf{h}}_i$ which consequently are used to calculate a forward and backward language model log-likelihood. Their model, ELMo, uses multiple layers of LSTMs and it has been pretrained on the Google Billion Word dataset. Another version of the model, ELMo 5.5B, has been trained on the English Wikipedia and monolingual news crawl data from WMT 2008-2012.

BERT: Instead of a standard language model objective, Devlin et al. (2018a) propose to sample positions in the input sequence randomly and to learn to fill the word at the masked position. To this end, they employ a Transformer architecture and train it on the BookCorpus (Zhu et al., 2015) as well as a crawl of English Wikipedia. In addition to this pseudo language model objective, they use an auxiliary binary classification objective to predict whether a particular sentence follows the given sequence of words.

3 Related Work

Many studies have investigated pretrained word representations, sentence representations, and language models. Existing work focuses on understanding linguistic and semantic properties of word representations or how well pretrained sentence representations and language models transfer linguistic knowledge to downstream tasks. In contrast, our investigation seeks to answer to what extent pretrained language models store factual and commonsense knowledge by comparing them with symbolic knowledge bases populated by traditional relation extraction approaches.

Baroni et al. (2014) present a systematic comparative analysis between neural word representation methods and more traditional count-based distributional semantic methods on lexical semantics tasks like semantic relatedness and concept categorization. They find that neural word representations outperform count-based distributional methods on the majority of the considered tasks. Hill et al. (2015) investigate to what degree word representations capture semantic meaning as measured by similarity between word pairs.

Marvin and Linzen (2018) assess the grammaticality of pretrained language models. Their dataset consists of sentence pairs with a grammatical and an ungrammatical sentence. While a good language model should assign higher probability to the grammatical sentence, they find that LSTMs do not learn syntax well.

Another line of work investigates the ability of pretrained sentence and language models to transfer knowledge to downstream natural language understanding tasks (Wang et al., 2018). While such an analysis sheds light on the transfer-learning

[2]Contextual representation models (Tenney et al., 2019) might be a better name, but we keep calling them language models for simplicity.

abilities of pretrained models for understanding short pieces of text, it provides little insight into whether these models can compete with traditional approaches to representing knowledge like symbolic knowledge bases.

More recently, McCoy et al. (2019) found that for natural language inference, a model based on BERT learns to rely heavily on fallible syntactic heuristics instead of a deeper understanding of the natural language input. Peters et al. (2018b) found that lower layers in ELMo specialize on local syntactic relationships, while higher layers can learn to model long-range relationships. Similarly, Goldberg (2019) found that BERT captures English syntactic phenomena remarkably well. Tenney et al. (2019) investigate to what extent language models encode sentence structure for different syntactic and semantic phenomena and found that they excel for the former but only provide small improvements for tasks that fall into the latter category. While this provides insights into the linguistic knowledge of language models, it does not provide insights into their factual and commonsense knowledge.

Radford et al. (2018) introduce a pretrained language model based on the Transformer which they termed generative pretraining (GPTv1). The first version of GPT (Radford et al., 2018) has been trained on the Book Corpus (Zhu et al., 2015) containing 7000 books. The closest to our investigation is the work by Radford et al. (2019) which introduces GPTv2 and investigates how well their language model does zero-shot transfer to a range of downstream tasks. They find that GPTv2 achieves an F_1 of 55 for answering questions in CoQA (Reddy et al., 2018) and 4.1% accuracy on the Natural Questions dataset (Kwiatkowski et al., 2019), in both cases without making use of annotated question-answer pairs or an information retrieval step. While these results are encouraging and hint at the ability of very large pretrained language models to memorize factual knowledge, the large GPTv2 model has not been made public and the publicly available small version achieves less than 1% on Natural Questions (5.3 times worse than the large model). Thus, we decided to not include GPTv2 in our study. Similarly, we do not include GPTv1 in this study as it uses a limited lower-cased vocabulary, making it incompatible to the way we assess the other language models.

4 The LAMA Probe

We introduce the *LAMA* (LAnguage Model Analysis) probe to test the factual and commonsense knowledge in language models. It provides a set of knowledge sources which are composed of a corpus of facts. Facts are either subject-relation-object triples or question-answer pairs. Each fact is converted into a cloze statement which is used to query the language model for a missing token. We evaluate each model based on how highly it ranks the ground truth token against every other word in a fixed candidate vocabulary. This is similar to ranking-based metrics from the knowledge base completion literature (Bordes et al., 2013; Nickel et al., 2016). Our assumption is that models which rank ground truth tokens high for these cloze statements have more factual knowledge. We discuss each step in detail next and provide considerations on the probe below.

4.1 Knowledge Sources

To assess the different language models in Section 2, we cover a variety of sources of factual and commonsense knowledge. For each source, we describe the origin of fact triples (or question-answer pairs), how we transform them into cloze templates, and to what extent aligned texts exist in Wikipedia that are known to express a particular fact. We use the latter information in supervised baselines that extract knowledge representations directly from the aligned text.

4.1.1 Google-RE

The Google-RE corpus[3] contains ∼60K facts manually extracted from Wikipedia. It covers five relations but we consider only three of them, namely "place of birth", "date of birth" and "place of death". We exclude the other two because they contain mainly multi-tokens objects that are not supported in our evaluation. We manually define a template for each considered relation, *e.g.*, "[S] was born in [O]" for "place of birth". Each fact in the Google-RE dataset is, by design, manually aligned to a short piece of Wikipedia text supporting it.

4.1.2 T-REx

The T-REx knowledge source is a subset of Wikidata triples. It is derived from the T-REx

[3] `https://code.google.com/archive/p/relation-extraction-corpus/`